Ayush Jain Software Engineer

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SKILLS

Programming Languages: Java, Python, C++, SQL | **Databases & Vector Stores:** Databases & Vector Stores: PostgreSQL, MySQL, MongoDB, SQLite ChromaDB | **Backend Technologies:** Spring Boot, Spring Data JPA, Hibernate, RESTful APIs, Microservices | **Tools & Platforms:** Git, GitHub, Docker, Jenkins, Maven, Postman, Splunk, Grafana, Linux | **Architecture & Design:** System Design, Distributed Systems, MVC Architecture, Microservices

EXPERIENCE

Tata Consultancy Services (Client: American Express) 01/2025 – Present | Bengaluru, India
Software Engineer

- Developed and enhanced scalable Spring Boot REST APIs for multi-rail settlement systems supporting 1,000+ daily transactions, enabling reliable batch-based payment processing while maintaining clean, scalable, and maintainable code.
- Performed root cause analysis (RCA) on authorization and payment-processing defects, partnering with engineering and business stakeholders to identify failure patterns and implement fixes that contributed to an estimated 8–10% reduction in transaction declines.
- Coordinated with 8+ cross-functional teams across the American Express payment network ecosystem to deliver features, troubleshoot integration issues, and support critical authorization and settlement workflows.
- Supported weekly production deployments annually through weekly releases and near-daily hotfixes in the E3 production environment, applying traffic management, deployment, and rollback strategies to maintain system availability.
- Leveraged Splunk, Kibana, and Postman to monitor distributed applications, diagnose production incidents, and support large-scale payment systems processing 1K+ daily transactions

PROJECTS

IntelliERP – AI-Powered Business Operations System
Python/FastAPI, MongoDB storage, React frontend, AI integration using Gemini/Claude/OpenAI function-calling

- Built an end-to-end ERP system for Indian SMEs covering sales, purchase, inventory, accounting, GST, and multi-branch management with dashboards and reporting features.
- Developed AI-powered automation features for invoice scanning, product onboarding, duplicate detection, and conversational business assistance integrated into ERP workflows.
- Designed a FastAPI-based backend AI pipeline integrating LLMs (OpenAI, Gemini, Claude) with OCR tools (pdfplumber, PyMuPDF, pytesseract) for invoice parsing and structured data extraction.
- Improved operational efficiency by reducing manual entry, enabling accurate supplier/product matching, and automating invoice-to-bill conversion and purchase workflows.

CartPilot – AI Shopping Copilot
Python, LangGraph, LangChain, OpenAI GPT-4, SQLite

- Built an agentic AI shopping assistant capable of understanding natural language queries, planning multi-step actions, and autonomously retrieving relevant product information, reducing manual search effort by 80%+.
- Designed a multi-agent workflow integrating product search, review analysis, recommendation, and reasoning modules to deliver personalized purchase suggestions with sub-second tool orchestration.
- Implemented tool-calling agents to query structured product and review databases, enabling analysis of 1,000+ products and generating context-aware recommendations based on ratings, price, and user preferences.
- Engineered a modular, extensible agent architecture with persistent memory and structured outputs, improving recommendation relevance by 30% while supporting seamless integration of additional tools and APIs.

EDUCATION

IIIT Bhubaneswar 2020 – 2024 | Bhubaneswar
Bachelor of Technology, Computer Engineering

CERTIFICATES AND ACHIVEMENTS

- Achieved **Knight** rating on LeetCode and solved **500+ algorithmic problems**, strengthening expertise in Data Structures, Algorithms, Problem Solving, and Competitive Programming.
- Finalist at HackNITR 3.0, building innovative software solutions and collaborating in a high-pressure competitive development environment.